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Introduction to Remote Sensing

Lab_08a: Calculating NDVI & NDWI Indices Using QGIS

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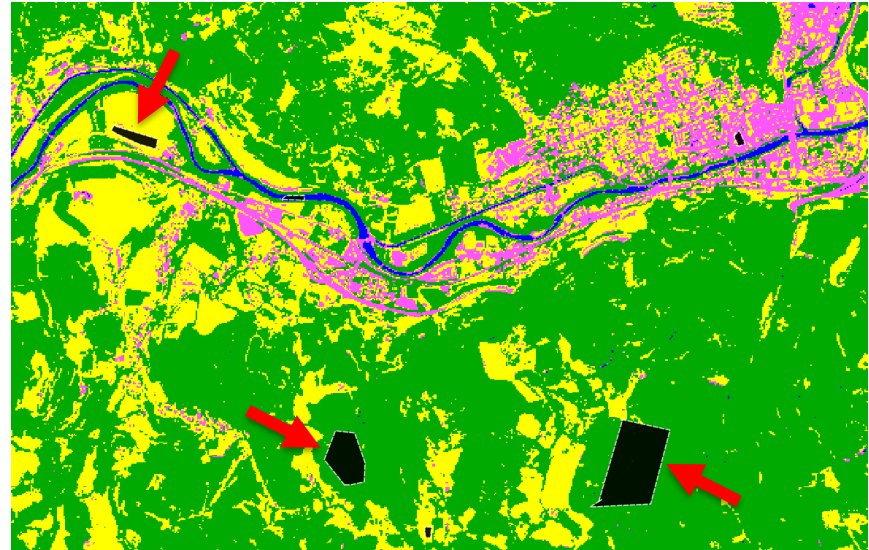
402.467 / Applied GIS
Fundamentals

Refresh our memories

Sentinel-2 image tile of Graz

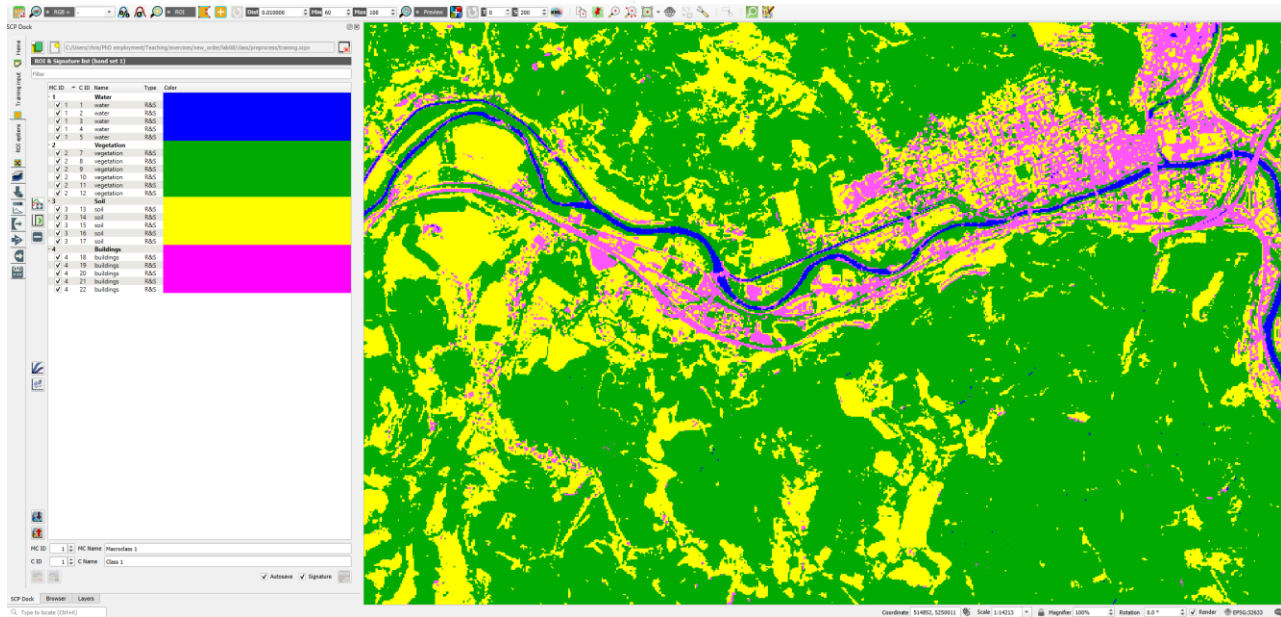


Supervised image classification (ROIs)



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Fundamentals

Refresh our memories (1)



Semi-Automatic Classification Plugin

Plugin ID: 284

The Semi-Automatic Classification Plugin (SCP) allows for the supervised classification of remote sensing images, providing tools for the download, the preprocessing and postprocessing of images.

★★★★★ (241) votes

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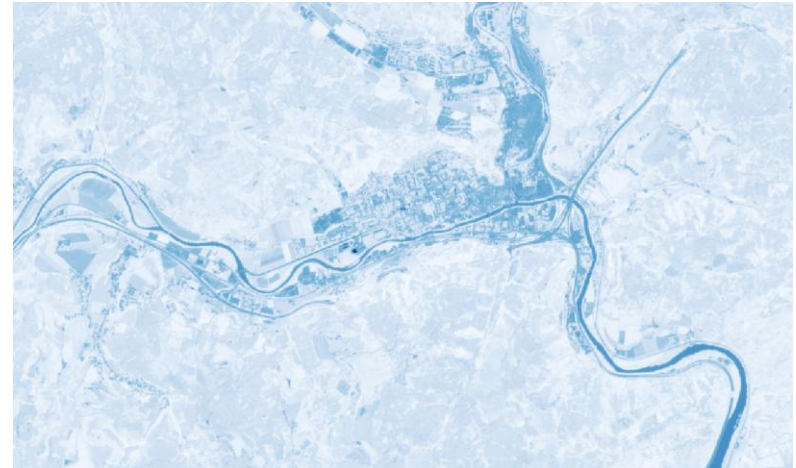
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1	1	Water	RGB	Blue
1	2	Water	RGB	Blue
1	3	Water	RGB	Blue
1	4	Water	RGB	Blue
1	5	Water	RGB	Blue
2	1	Vegetation	RGB	Green
2	2	Vegetation	RGB	Green
2	3	Vegetation	RGB	Green
2	4	Vegetation	RGB	Green
2	5	Vegetation	RGB	Green
2	6	Vegetation	RGB	Green
2	7	Vegetation	RGB	Green
2	8	Vegetation	RGB	Green
2	9	Vegetation	RGB	Green
2	10	Vegetation	RGB	Green
2	11	Vegetation	RGB	Green
2	12	Vegetation	RGB	Green
3	13	Soil	RGB	Yellow
3	14	Soil	RGB	Yellow
3	15	Soil	RGB	Yellow
3	16	Soil	RGB	Yellow
3	17	Soil	RGB	Yellow
4	18	Buildings	RGB	Magenta
4	19	Buildings	RGB	Magenta
4	20	Buildings	RGB	Magenta
4	21	Buildings	RGB	Magenta
4	22	Buildings	RGB	Magenta

Intro to Remote Sensing Indices

Normalized Difference Vegetation Index (NDVI)



Normalized Difference Water Index (NDWI)



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Fundamentals

NDVI

- Is used to assess **crop health** from satellite images
- NDVI **values** run from **-1 to +1**, where **negative** values suggest **non-vegetated areas** & **positive** values suggest **healthy vegetation**
- Is calculated using the **form**:

$$NDVI = \frac{(NIR - RED)}{(NIR + RED)}$$

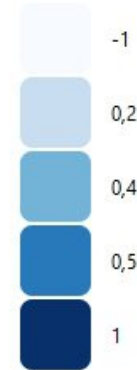
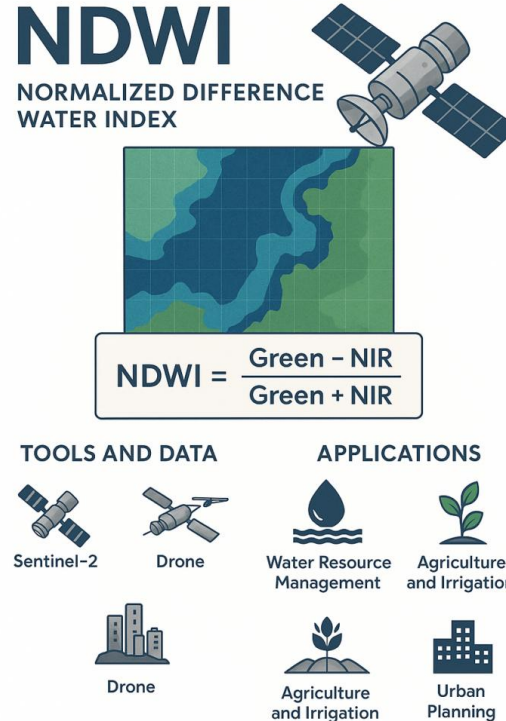


NDWI

- Is used to **highlight open water features** in satellite images, allowing a water body to *stand out* against the soil or vegetation
- Water bodies** typically have **positive** NDWI **values**, while vegetation and soil tend to be lower or negative
- Is calculated using the **form**:

$$NDWI = \frac{(Green - NIR)}{(Green + NIR)}$$

by McFeeters (1996)



Group projects

By next week, have an initial topic idea & send me by email your group members with an indicative title

